THE STORY BEHIND THE EXPERIMENT

The curled torus burps

A small push, a large release,
and the question of what survives

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Everything you have ever seen happened in space with three large directions you can move in. We know its rules well. What we do not know is how it came to be.

This paper tests one piece of one hypothesis about that: a sharp change from something else into space. In a small world of dots and links it asks three questions. What does it take to start such a change? What comes out when it happens? What is left behind?

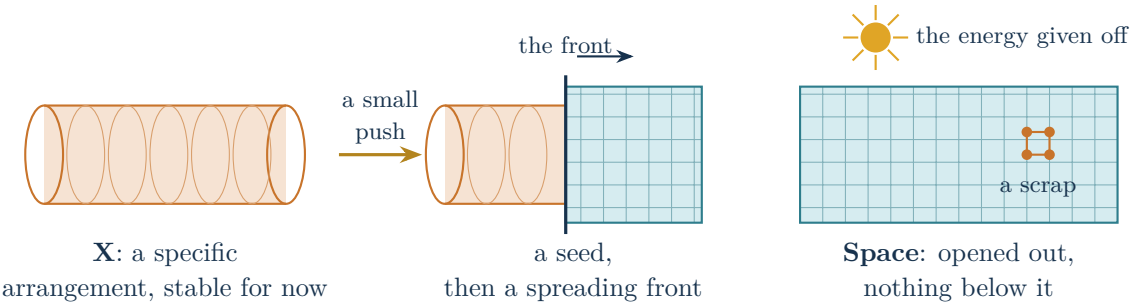


Figure 1. The idea in one picture. SKETCH, NOT DATA What came before space, called X here (left), waits until a small push starts it; the new arrangement spreads behind a front (middle); when it is done, space holds the energy that came out plus a small scrap of the old arrangement (right). Everything in this paper tests some part of this picture inside one small model world. The single front and the large opened space are the hypothesis, not outcomes of these runs. Chapter 3 says why X is drawn as a tube and what both shapes are in that world, where each closes on itself; Chapter 6 shows what the runs reached.

Who is saying what

The project keeps four kinds of statement apart, and this edition tags them where it matters, so that a sentence is never quietly promoted from one kind to another. EXACT is arithmetic, or a complete listing of every possibility: true whatever a run does. MEASURED is a simulation whose rules and predictions were written down before it ran, with its verdict; MEASURED, EXPLORATORY is a run that was not formally pre-registered, or a closer look at runs already read. OUR READING, UNVERIFIED is a reading of a result, offered by the author and her AI assistants and checked by no physicist. THE AUTHOR'S CLAIM is the author's hypothesis about reality, stated as hers. One more tag, PUBLISHED, marks other researchers' published results. Numbers in square brackets point to the sources listed at the end.

1 THE PLACE WE START

We live in space, and we cannot rerun its birth

The author. Emily Smith is a software engineer, not a physicist, and also a startup founder and a foster mom, so her recreational time is scarce and the computing bills are her own. The project is called RecreationalPhysics anyway. Her reason is simple: she wants to map her intuition about how reality is organized onto what humanity knows about the universe.

Her hypothesis. THE AUTHOR'S CLAIM Space is one settled arrangement of something deeper, which she calls X because nobody knows what it is. X was not chaos. It was a specific arrangement, stable for now, the way a reagent is stable until something gives it a push. The change that made space started at a seed, spread as a front, released a definite amount of energy all at once, which became matter and light, and left a remainder. Across the change the energy books balance. Figure 1 is that paragraph as a picture.

The approach. That is a hypothesis, not a result, and nothing in a computer can prove it about reality. What a computer can do is take a small world whose rules are written out in full, in which space is an arrangement and not a given, and ask whether a change of this kind can happen there at all. If the answer were no, the idea would have no home in that world. If it is yes, the idea gains a laboratory, and every number in it becomes a thing to check. Three habits keep the laboratory honest: published results are reproduced first, each prediction is written down before its run, and a result that goes against the author is reported as prominently as one that goes her way.

The parts of this paper. Chapter 2 is the starting point: a published model, what we reproduced of it, and the one thing we change. Chapters 3 to 9 are the experiment. They take one specific arrangement and ask what it is, what it costs to start changing, how long it waits, what takes shape, where the energy goes, what remains, and whether the answers hold at other settings. Chapters 10 and 11 say where the simulations stop, what held, what failed, and what the next papers take up.

A toy, and why this one. Physicists call a small world of that kind a **toy model**: simple enough that every consequence of its rules can be counted or simulated, and not to be mistaken for the real thing. Ours starts from a published one, combinatorial quantum gravity [1, 2, 3], for four reasons. Space is not built into it: there are only dots and links, and geometry has to form. Its energy is a version of the curvature in Einstein's theory of gravity. It has published results to check our code against. And at the smallest sizes every possible arrangement can be listed, which gives exact answers to test the simulations on.

2 THE TOY

A published model, reproduced, and the one knob we add

PUBLISHED Combinatorial quantum gravity is the work of Carlo Trugenberger, Christy Kelly and Fabio Biancalana [1, 2, 3]. It is a network of dots and links with no positions at all: a dot has no place, only neighbors. Mathematicians call such a thing a **graph**. Figure 2 is the whole of it.

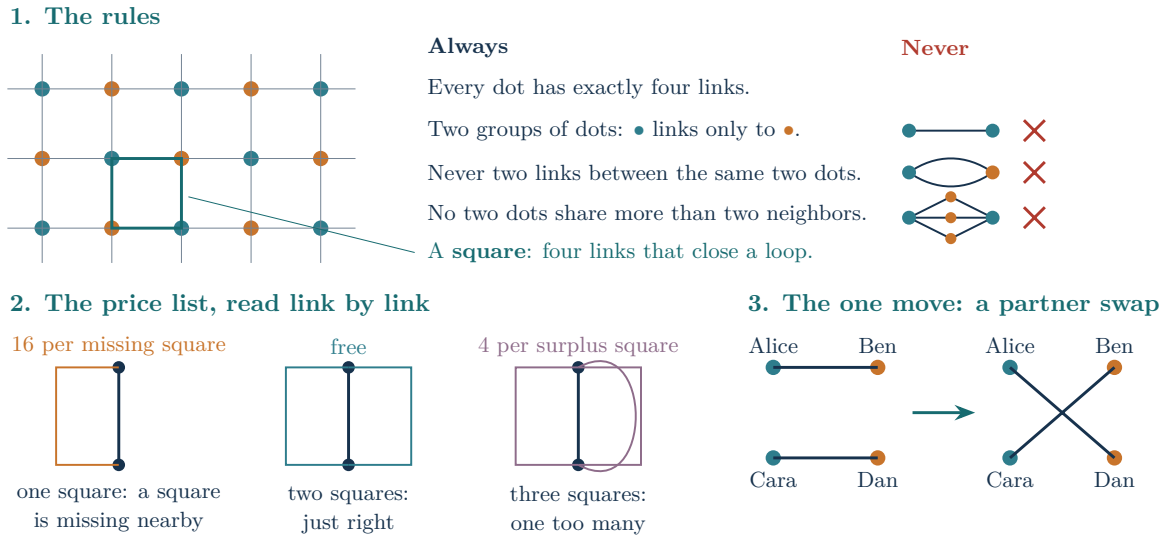


Figure 2. The published model on one card. PUBLISHED There are no positions: drawing the dots on a grid is a convenience, and only the links are real. The energy is worked out from the **squares**: a link on exactly two is free, fewer means a square is missing nearby, and a third is a **surplus square**. In a swap every dot keeps its four links and its group, and a swap that would break a rule is thrown out.

PUBLISHED With those two prices the energy measures how curved the network is: a graph version of the curvature in Einstein's equations of gravity. It has no meters or joules in it. We call its amount a **unit**, and leave the conversion into physical energy for a later paper. The published work also studies a simpler version with the surplus price switched off, where a third square is free.

PUBLISHED A run is a long chain of swaps, each one proposed at random. A swap that lowers the energy is always accepted. A swap that raises it by ΔE units is accepted with a chance of $e^{-\Delta E/g}$, where e is a fixed number, about 2.7, and g plays the part of temperature. We call g **the warmth**: the bigger the cost, the rarer the swap, and the warmer the network, the more often a costly swap goes through. Picture an endless **bath** that lends energy for an uphill swap and takes back what a downhill swap gives off. Time is counted in **sweeps**: one sweep is two proposed swaps for every dot. This is the standard way to sample such a model, not a claim about how anything moved in real time.

Reproduced before changed

PUBLISHED The published work starts from a random tangle of links, cools it, and asks how geometry condenses out of the tangle. Sharply, the way water freezes, giving off heat with ice and water side by side? (Very clean water can even stay liquid below zero, stuck, until a jolt sets it off.) Or smoothly, the way a fridge magnet loses its pull as it warms, with no burst of heat and no moment when half of it has changed? Physicists call the two kinds **first order** and **continuous**. Published work reports

continuous behavior for curvature-based models [2, 3] and first-order behavior in a related model with no surplus price [4], where the network breaks into closed pieces.

MEASURED We checked published results before extending the model.[1] Our simulations match the 2019 square-density curve [2] with a root-mean-square difference of 0.005 at the same size, match exact averages at sizes small enough to list every possible arrangement, and reproduce the sharp change into closed pieces with the surplus price off. Then, with the rules written down first, we measured the jump: the energy given off per dot as the tangle orders.

Price list	Energy given off per dot, in units, as the tangle orders
Surplus price off	10.3, 11.3 and 12.5 at 48, 64 and 96 dots: large, and growing
Full, as published	at most 1.65, 1.51 and 1.30 at 36, 64 and 100 dots: small, and falling

MEASURED With the surplus price off the jump is large, but what forms is a pile of closed pieces, not a space. With the full price list we found no clear jump at any size we ran. That settles nothing, since a small network can hide a jump a larger one would show: the registered tests there were inconclusive.

The one thing we change

The toy turns one price into a knob. In the toy a surplus square costs 4λ units, where λ (lambda) is a number that can be turned up or down, so we call it **the knob**. With the knob at exactly $\lambda = 1$ the toy is the published model. With the knob at $\lambda = 0$ it is the published simpler version. Every other setting is our addition, and there the energy is still a perfectly good rule for a network, but it is no longer a curvature. So two names are kept apart from here on. **The published model** is theirs. **The toy** is ours: a close cousin of combinatorial quantum gravity, not the thing itself.

We have not found published work that sets the surplus price to any value above 1 [13]. In print, as far as we have found, are the two settings just described and, in the text of two of the papers [2, 14], a version in which a third square is forbidden outright. So, barring a gap in our search, the approach taken here is new.

Why turn the knob at all? MEASURED Not to settle sharp or smooth: out of a random tangle, settings of 1.25 and 1.5 gave the same picture as 1. Neither this paper nor the author's idea rests on that test. What she asks of the toy is plainer: does a prepared arrangement give way with a jump in energy at the sizes run, and how large is the jump? Chapter 3 shows how the knob makes room for that question.

The point to keep: The published model is a network with a price list, 16 per missing square and 4 per surplus square, which makes its energy a curvature. We reproduced it before changing anything; with the full price list our sizes show no clear jump out of randomness, which settles nothing either way. The toy changes one thing: a surplus square costs 4λ, and only $\lambda = 1$ is the published model.

[1] The published simulations are not identical experiments: the 2019 simulation forbids surplus squares, the later curvature action permits them, and the study with no surplus price does not use the same fixed two-group networks. Our square-density curve differs from a 2025 figure [3] by up to 0.41 over part of its range. Given the different stated protocols, agreement with the 2019 curve does not validate our whole curve, and this reproduction issue remains open. Kelly's doctoral thesis [5] adds a hint for four links per dot: cooled and then warmed again, the network does not quite retrace its path, though the thesis declines to call the evidence definitive.

3 THE PHASE BEFORE

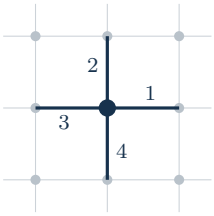
What if what came before space was not random?

The published question starts from a random tangle. This program asks a different one: what if what came before space was not random at all, but a specific arrangement, stable for now, with energy to give? To ask it in the toy, we first need to say what space looks like in a network with no positions, and what the alternative to it could be.

Directions

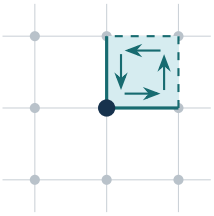
Space, at its simplest, is a set of directions with room in them. Figure 3 shows how to find a direction in a network that has no positions.

1. Four links, six pairs



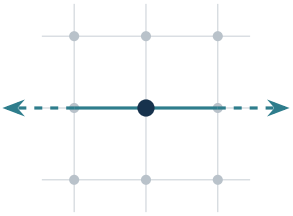
Every dot has four links. Four links make six pairs: 1–2, 2–3, 3–4, 4–1, 1–3 and 2–4.

2. Right angles: a square



A pair at right angles closes a square: out along one link, across, and back along the other. Four of the six pairs do this.

3. Straight through: a direction



A pair that runs straight through closes no square. That is a **direction**. Two of the six pairs do this.

Figure 3. What a direction is, in a network with no positions. EXACT Count the pairs at a dot that close no square, and call the count d. On a grid, $d = 2$.

A grid with no edges

A grid drawn on paper has edges, where the dots run out of links. The toy removes the edges.

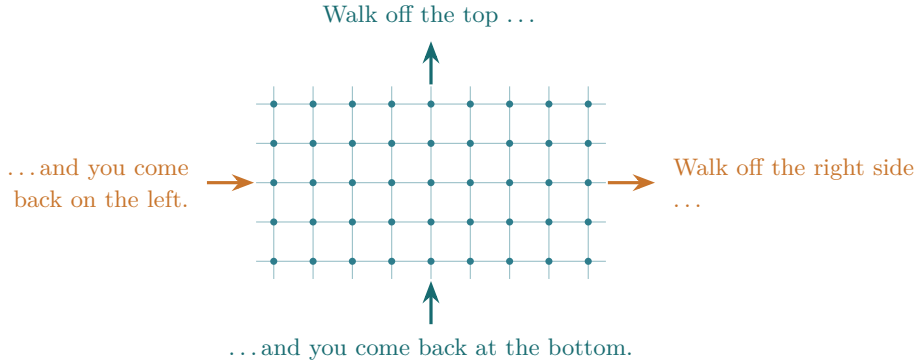


Figure 4. A grid with no edges. EXACT The grid wraps around both ways, like an old game screen whose edges join, so every dot has its four links, and when both ways around are long every dot reads $d = 2$. We call it **the flat sheet**. Mathematicians call any grid that wraps both ways a **torus**, the surface of a doughnut; arbitrary networks reached during a run need not be one.

Curling

If a direction is not large, what else can it be? Small: curled up on itself.

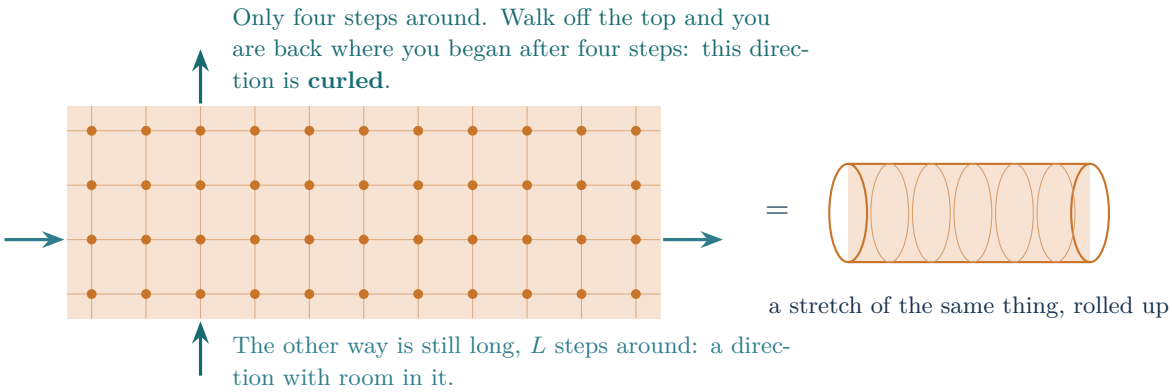


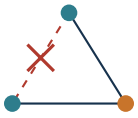
Figure 5. Curling one direction. EXACT Make one of the two ways around only four steps long. A grid four dots around and L dots along, wrapped both ways, is still a torus, the $4 \times L$ torus: the curled kind. We call it **the tube**. Its long way wraps around too, so the tube has no open ends.

Two steps around?



No. That would take two links between the same two dots, which the rules forbid.

Three steps around?



No. The last link would join two dots of the same group. Links only join the two groups, so every closed walk takes an even number of steps.

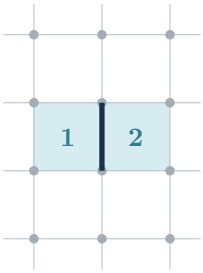
Four steps around?



Yes, and it is the tightest curl there is. The walk around is itself a square.

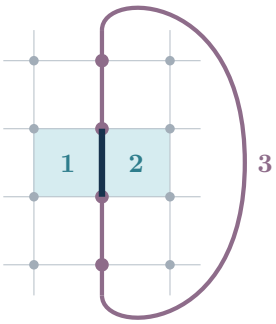
Figure 6. Why four steps around, and not three or two. EXACT

On the flat sheet



The thick link borders two squares, one on each side. Two is free.

On the tube



1 and 2: the squares on either side, as before.

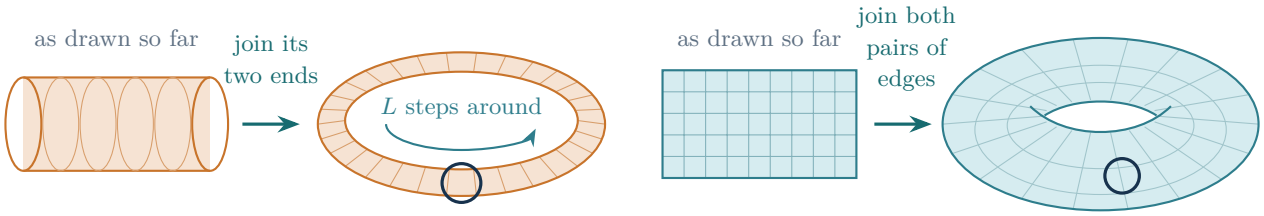
3: the ring of four links around the tube, its top joined to its bottom. It is a square too, and the thick link is one of its sides.

Three squares on one link is one too many. The third is a **surplus square**, and the price list charges 4λ for it.

Figure 7. Curling puts a third square on a link. EXACT So curling a direction is exactly what the surplus price charges for. And because the way around is now a square, the pair of links that points that way closes one: the direction no longer counts, and every dot on the tube reads $d = 1$.

Ring and doughnut: flat is not the same as open

The drawings so far leave something out: the edges are joined. EXACT Join the tube's two ends and it is a thin ring, like a hula hoop, four steps around the thin way and L steps around the long way. The sheet is a fatter doughnut, long both ways (Figure 8). Nothing in the toy is open-ended, and nothing can be: every dot must have its four links, so there is no edge to stop at, and with a finite number of dots a walk that keeps going must come back around.



The tube is a thin ring. Four steps around the thin way. What an inhabitant can reach (the circle) goes right around it.

The sheet is a fatter doughnut. Long both ways. What an inhabitant can reach (the circle) is a patch that looks like an endless grid.

Figure 8. The two arrangements with their edges joined. EXACT Neither has an end or an edge. SKETCH, NOT DATA Drawn in three dimensions the doughnut looks bent. The network is not: it has no positions, only links, and the game-screen picture of Figure 4 keeps its distances true.

Two words are kept apart from here on. **Flat** is about each dot's surroundings: every link on exactly two squares, the wiring of a plain grid, at zero energy. **Wide** is about the whole shape: how many steps it takes to walk around it and come back. A doughnut from a bakery is not flat, since its skin is stretched on the outside and squeezed around the hole, but the toy's doughnut is the game-screen kind and is not bent anywhere. The sheet is flat and wide. The tube is neither.

Can anyone inside tell a wide doughnut from an endless grid? Suppose something lived in the network and could learn only what it reaches by walking links. Let it count the dots within one step of home, then two, then three. EXACT On a wrapping grid the counts match an endless grid's until walks that set out in opposite directions meet around the back, and then they fall short. In the table each grid is long the other way, and bold marks the first count that falls short.

Dots within	1 step	2 steps	3 steps	4 steps	6 steps
An endless grid	5	13	25	41	85
The tube: 4 steps around	5	**12**	20	28	44
A grid 8 steps around	5	13	25	**40**	72
A grid 12 steps around	5	13	25	41	**84**

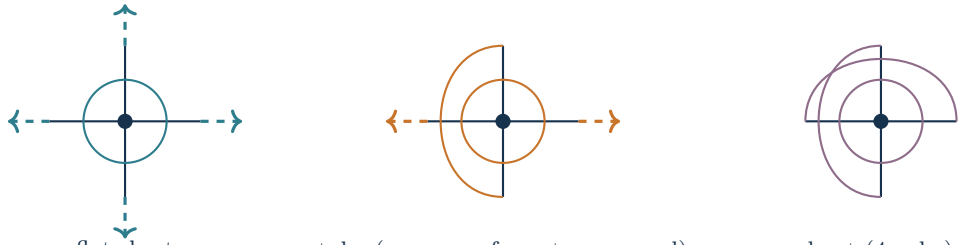
EXACT The count first falls short at half the way around. So from inside, a doughnut much wider than its inhabitants is open in every way they can check. The tube is the opposite case. Its count falls short at two steps, as early as a count can, because its way around is the smallest loop the rules allow. That direction has no room in it at all, which is what $d = 1$ records: from inside, the tube is little more than a line.

PUBLISHED Astronomers put the same question to real space. On the largest scales, observations are consistent with zero spatial curvature within the cosmological analysis used to interpret them [12], though gravity still curves spacetime around stars and planets. Whether space is also endless is a separate question. A search of the sky for the repeated patterns that a space closed up like a doughnut would show found none on scales smaller than the part of space we can see [15]. Flat has been measured. Endless has not.

The knot, and the count for all three

Curl both ways around to four steps and every dot reads $d = 0$: a closed sixteen-dot knot called the 4-cube. It is a torus too, curled both ways. Most of the closed pieces of Chapter 2 are these knots. Figure 9 puts the three arrangements side by side.

One dot in each arrangement. An arc marks a pair of links that closes a square. A dashed arrow marks a pair that closes no square: a direction. A loop marks a pair that is curled shut.

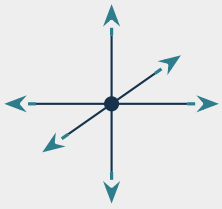


flat sheet
$d = 2$

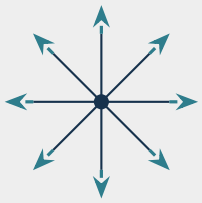
tube (one wrap four steps around)
$d = 1$

knot (4-cube)
$d = 0$

Figure 9. How many directions a dot has. EXACT With four links per dot, the count d is the number of pairs of a dot's links that close no square, and it tells the flat sheet, the tube and the knot apart dot by dot. It is a signature, not a certified geometry: Chapter 6 says what it can and cannot vouch for. This paper is about the simplest change among the three: one curled direction opening, d from 1 to 2.



six links per dot: 15 pairs; 12 at right angles close squares, 3 run straight through a flat grid has d=3



eight links per dot: 28 pairs; 24 at right angles close squares, 4 run straight through a flat grid has d=4 (drawn flat: paper has only two)

Outside this paper.
Six and eight links per dot give three and four directions. A later paper takes them up. Nothing in this paper depends on them.

Figure 10. Outside this paper. EXACT The same count with six or eight links per dot allows three or four directions, the subject of a later paper.

The ladder

EXACT Put the three arrangements side by side: flat sheet, tube, knot. All three are tori, and we tell them apart by how they are curled: flat, curled, knot. Each curls one more direction than the one before, so we call them the rungs of a **ladder**. The price list of Chapter 2 gives each rung its bill, per dot:

	flat sheet	tube	knot
Directions curled	none	one	both
Squares per dot	1	$1\frac{1}{4}$	$1\frac{1}{2}$
saved on squares, at 16 each	0	4	8
Surplus squares per dot	0	1	2
paid for surplus, at 4λ each	0	4λ	8λ
Bill per dot: paid minus saved	**0**	$\mathbf{4(\lambda-1)}$	$\mathbf{8(\lambda-1)}$
at $\lambda = 1.25$, where a surplus square costs 5	0	1	2

Curling is a trade between the squares it gains and the surplus it pays for, and the knob decides which side wins.

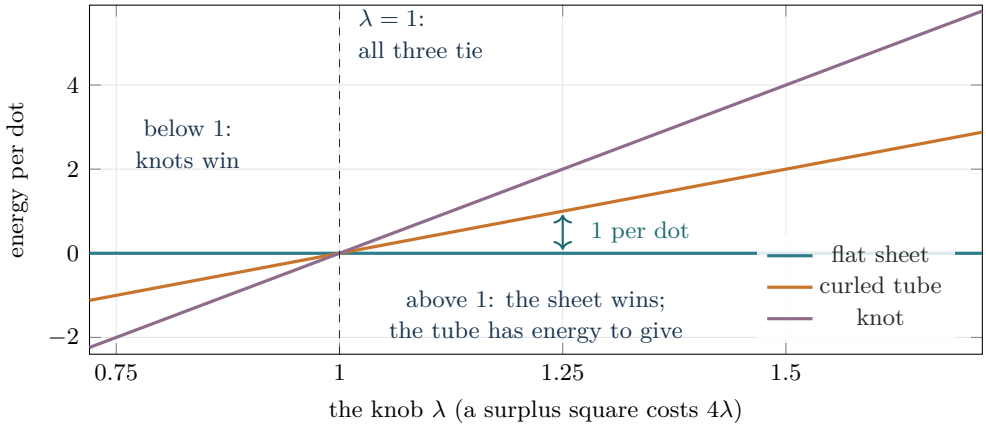


Figure 11. The ladder against the knob. EXACT Each line is one rung's energy per dot as the surplus price changes. At $\lambda = 1$, the published setting, the three rungs cost the same, so turning one into another releases nothing. Above 1 the sheet is lowest and the tube sits above it with a known amount to give: at $\lambda = 1.25$, exactly one unit per dot. Below 1 the ladder flips and a sheet would gain by curling into knots, which is why the network breaks into knots when the surplus price is off.

This is what the knob is for. EXACT At the published setting, curled and flat cost the same, so turning one into the other gives off nothing. Equal cost does not make the two equally likely, and it does not forbid a change. Turn the knob above 1, and a tube that ends at zero energy gives off exactly $4(\lambda - 1)$ units per dot. That is a reason to study this cousin of the published model. It is not proof of a first-order transition, and it is not a reason why nature should choose it.

That release is what the rest of this paper is about, so it gets a name of its own. The author calls it **the burp**: the energy given off, and by extension the event that gives it off, a quiet wait, a start, a

spreading change and a definite release.

PUBLISHED The standard account of a stable-for-now state giving way calls its central calculation "the bounce" [8]. The two names point at opposite ends of the event. The bounce is about the start: it gives how often a seed of the new state appears, in a quantum system that tunnels *through* its hill. The burp is about the end: what comes out. And the toy has no tunneling. Its tube goes *over* the rim, with energy lent by the bath or handed to it as a push, never through it.

EXACT The changed rule is the published curvature energy plus $4(\lambda-1)$ times the surplus square count. The extra term vanishes when that count is zero. THE AUTHOR'S CLAIM The author proposes reading it as a *curling cost*: energy has to be paid to curl space tight, and is given back when it uncurls. She holds that as something to test. EXACT In the toy it is plain bookkeeping: a curled tube sits $4(\lambda-1)$ per dot above the flat arrangement, so uncurling gives that much back and curling again would have to pay it. OUR READING, UNVERIFIED What is not supplied is a derivation. At $\lambda = 1$ the rule is the published graph version of the curvature in Einstein's gravity; nothing known requires the extra term or fixes its size, so its meaning for nature is unverified.

Stable for now: is there a window?

Energy to give is not enough. The clean water of Chapter 2, still liquid below zero, has energy to give and sits there. For the tube to be a stand-in for X it must be stuck: leaving its equal-energy family must cost energy. EXACT At 18 dots the toy is small enough to list every allowed arrangement: there are 26, counting arrangements that differ only in the names of their dots as one. Among them, arrangements above the flat sheet from which every single swap costs energy exist for knob settings between 1 and 1.6, and for no larger setting. So the knob has a window, and a narrow one.

The stand-in

Our stand-in for X is therefore the tube. At $\lambda = 1.25$ it sits exactly one unit per dot above the flat sheet. It is a stand-in and not X: it has one direction with room in it where space has three, its curled direction is always four dots around, and it lives in a network of a few hundred dots. What it shares with X is the shape of the idea under test: a specific, curled, stable-for-now arrangement with energy to give.

The running example

Take a tube of 96 dots: four around, 24 along. At $\lambda = 1.25$ it holds 96 units above the flat sheet. That is the burp on offer. We will follow this tube through the rest of the paper so that the amounts stay concrete. It is a running example, not an invented history: the evidence comes from hundreds of repeated runs and, where possible, from counting every allowed move.

Now the question that decides whether any of this is a story at all. Ninety-six units are waiting to come out. What does it cost to start?

The point to keep: A direction, in the toy, is a pair of links at a dot that closes no square. Curling a direction four steps around turns the walk around it into a square, which puts a surplus square on its links. The tube has one direction with room in it where the sheet has two, and sits $4(\lambda - 1)$ per dot above it: one unit per dot at $\lambda = 1.25$. Both close on themselves: the tube is a thin ring and the sheet a fatter doughnut, flat at every dot and, from inside, as good as endless to anything much smaller than the way around it. At $\lambda = 1$ there is nothing to release, which is why the knob sits above 1, and why the toy is a cousin of the published model and not that model itself.

4 THE OBSTACLE

What does it cost to start?

The experiment: the arrangement • **the cost** • the wait • the change • the energy • what remains • other settings

Before any number, consider what the number could mean. If the push needed were tiny, a few units, then any warmth at all would set the tube off, and it would not be stable for now. If the push were as large as the burp itself, 96 units for our tube and more for a bigger one, then starting the change would cost what the change gives back, and the cost would grow with the size of the tube. Nothing small could ever start something large. A sharp change of the kind pictured at the start of this paper needs a middle answer: a push that is small, and that does not grow with the size of the thing being pushed, so that a bigger tube releases a bigger burp from the same spark. A fixed push and a growing release is the condition for a runaway.

So: is the cheapest exit from a perfect tube small, fixed, or neither?

Pricing the cheapest exit

EXACT Start from a perfect tube and try every swap the rules allow, the one move of Chapter 2. Each one breaks some squares and removes some surplus, and the price list gives its cost. Most are expensive. Some cost nothing: they twist the tube around itself without changing any dot's neighborhood, so they do not count as leaving. Among the swaps that do leave, two kinds are cheapest. Call them route A and route B. Here are their bills at $\lambda = 1.25$, where a surplus square costs 5:

	Route A	Route B
Squares broken	2	4
charged, at 16 each	+32	+64
Surplus squares removed	4	10
refunded, at 5 each	−20	−50
The bill	**12**	**14**

Route A is the cheaper at this setting; the two tie at $\lambda = 4/3$, and B is the cheaper beyond. (For readers who want every exit listed with its formula, the technical paper [11] has them in Section II.)

The cheapest way out of the tube costs 12 units.

Twelve. Not 96, and not 2. For our 96-dot tube the push is an eighth of what comes out; for a 288-dot tube, which holds 288 units, the same 12 is a twenty-fourth. EXACT The cost of the cheapest single swap does not depend on how long the tube is, because a swap only touches a handful of neighboring dots; the burp does, because every dot contributes one unit. That is the middle answer the change needed, and it is arithmetic.

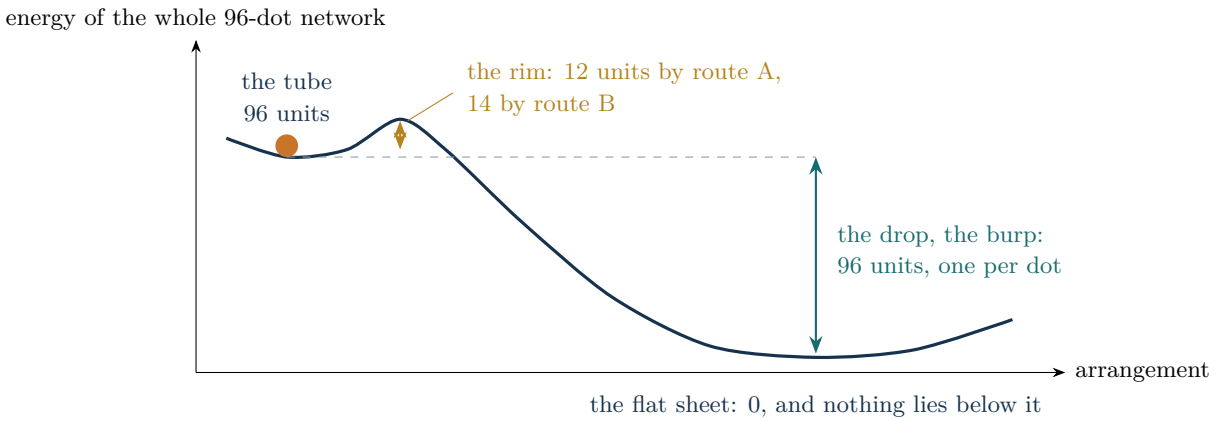


Figure 12. The change as a landscape, with its numbers. EXACT The hollow is the tube; the rim is the cheapest swap out; the floor is the flat sheet. For our running example the rim is 12 and the drop is 96. Make the tube three times longer and the rim is still 12 while the drop is 288.

Is 12 really the floor?

One cheap exit from the perfect tube is not the same as no cheaper exit anywhere nearby. The free twists lead to other arrangements with the same energy, and one of those might have a cheaper door. EXACT So the calculation followed every free twist, at six sizes from 48 to 288 dots, until no new arrangement appeared. At each size there are exactly three reachable arrangements, counting arrangements that differ only in the names of their dots as one; every one has the same menu of exits, and the cheapest costs 12. That is a complete listing at those sizes, not a proof for every size: at 32 dots the pattern is different.

The test from the other side

Arithmetic says nothing below 12 can leave. Does 12 suffice? To find out, take the bath away and hand the tube an exact budget and nothing more (Figure 13), a standard way to simulate at a fixed total energy [7].

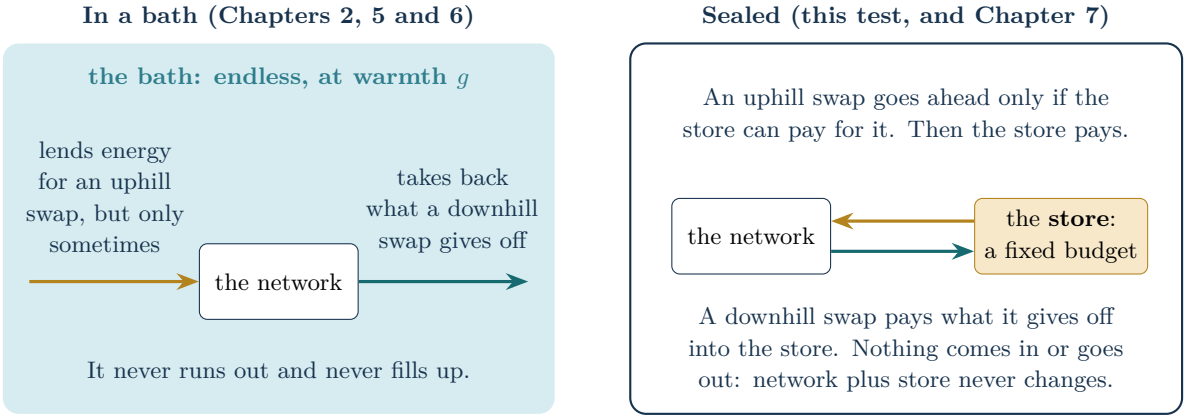


Figure 13. Two ways to run the toy. EXACT In a bath, energy for an uphill swap is always on offer, at a price in probability. **Sealed**, the network can spend only what its store holds. The store is not attached to any one place: it can pay for a swap anywhere in the network, so the push is a total budget, not a match held to one spot.

MEASURED, EXPLORATORY Try budgets of 4, 8, 10, 11, 11.5, 12, 13 and 16 units, eight runs at each budget at each of five sizes from 48 to 192 dots: 320 runs in all, each 12,000 sweeps long.

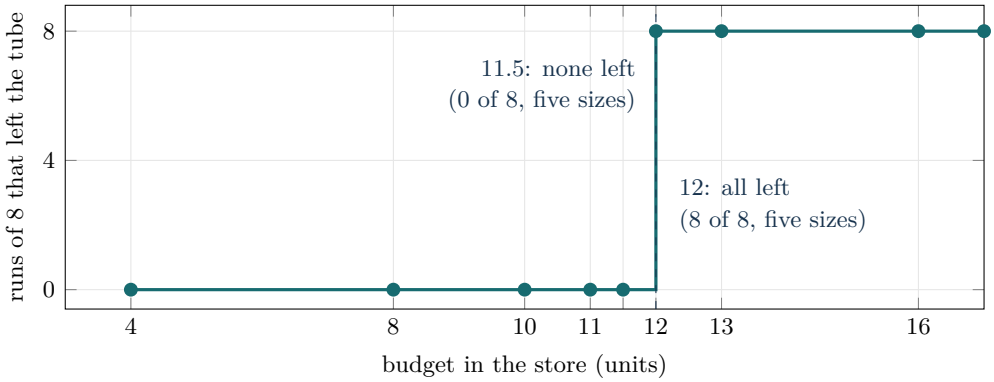


Figure 14. The staircase. MEASURED, EXPLORATORY The same curve at all five sizes, 48 to 192 dots: every budget below 12 gave zero escapes, every budget of 12 or more gave eight of eight. Pooled over the sizes, none of 200 runs left with a budget below 12, and all 120 left with 12 or more. The flag counted *leaving* the tube, not finishing the change.

EXACT The lower half of this staircase is a theorem of the toy at these sizes: with no exit cheaper than 12 and a store that cannot go negative, a budget below 12 can only pay for free twists. The upper half is a measurement. MEASURED, EXPLORATORY Together they say that the wall is local, exactly 12 units high at $\lambda = 1.25$, and the same at every size tried, while what waits behind it grows with the system.

Why we went here. This is the pivot of the whole paper. Suppose the push had grown with the size of the tube, so that a tube twice as long needed a push twice as large. Then no single spot could start the change: something would have to push the whole tube at once, and the picture of a seed and a spreading front would fail in this toy. We would have had to look for this kind of change in another one. A fixed, cheap push against a growing release is the condition under which something small can start something large.

The point to keep: Twelve units at $\lambda = 1.25$: the cost of the cheapest partner swap out of a perfect tube, counted exactly, the same whatever the tube's size, and confirmed from the other side by 320 sealed runs. The release grows with the tube; the push does not.

5 THE WAIT

How long before the door opens, and what kind of waiting is it?

The experiment: the arrangement • the cost • **the wait** • the change • the energy • what remains • other settings

A rim of 12 says what a push must be. It does not say when a tube left to itself will go. For that we put it back in the bath of Chapter 2 and let the network jiggle at warmth g. A swap that climbs the rim, costing 12, is accepted with chance $e^{-12/g}$: warmer, and the rim is cleared more often.

A lottery with counted tickets

Now the wait becomes a lottery, and every ticket can be counted. EXACT In one sweep the computer proposes two swaps for every dot, $2N$ in all for a tube of N dots, each one picked at random from every swap there is. How many of those proposals are exits?

Take route A. A tube of N dots has $3N$ different route-A swaps: three for every dot, so a longer tube has more ways out. But a longer tube also has more swaps of every other kind, and those grow faster, as N times N. So the chance that any one proposal lands on a route-A exit is 3 in $2N$, smaller for a longer tube. A sweep makes $2N$ proposals. Multiply the two and the length cancels:

$$2N \;\times\; \frac{3}{2N} \;=\; 3.$$

A sweep offers three route-A exits on average, whatever the size of the tube. A longer tube has more exits, each one is harder to hit, and with two proposals per dot in a sweep the two effects balance exactly. The same count for route B gives two.

Each offer of A is accepted with chance $e^{-12/g}$, each offer of B with chance $e^{-14/g}$. So the chance of leaving in any one sweep is $3\,e^{-12/g} + 2\,e^{-14/g}$, and the mean wait is one divided by that, with nothing fitted:

$$\text{mean wait} \;=\; \frac{1}{3\,e^{-12/g} + 2\,e^{-14/g}} \quad \text{sweeps.}$$

At $g = 1.5$, the warmth used for most runs below, that is 845 sweeps. Chemists know this shape as an Arrhenius law. Here both of its ingredients, the height of the rim and the number of tries per sweep, are counted from the arrangement rather than fitted to the data (the technical paper [11], Section II, gives the formula for any λ).

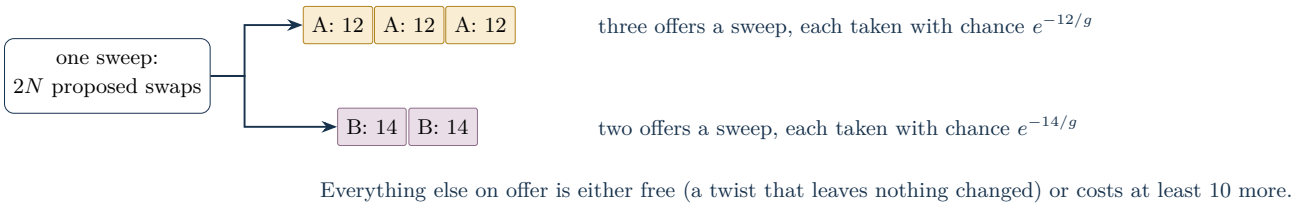


Figure 15. The lottery. EXACT The tickets are counted, not estimated: the perfect tube offers, on average, exactly three route-A exits and two route-B exits per sweep whatever its length. The chance of each being taken is set by the bath's warmth g.

Does the tube keep the appointment?

MEASURED, EXPLORATORY The first test ran twelve conditions at $\lambda = 1.25$: six warmths from $g = 1.4$ to 2.5, two sizes, sixteen tubes each, with measured waits from about 30 sweeps to about 2,500. The prediction written down for this test counted route A only, which makes the predicted wait a little longer than the full formula does: 994 sweeps at $g = 1.5$, where the full formula gives 845.

Averaged over the twelve conditions, the measured wait came within 2 percent of the predicted one. One condition at a time the match is looser: they range from 29 percent shorter than predicted to 62 percent longer (Figure 16). That is the scatter a lottery gives: with only sixteen tubes, their average wait lands, by chance alone, about 25 percent above or below the true average. The two sizes also differ. The 144-dot tubes sit near the full formula, and the 64-dot tubes waited longer. This exploratory test cannot say whether that is chance, which is one reason for the stricter test below.

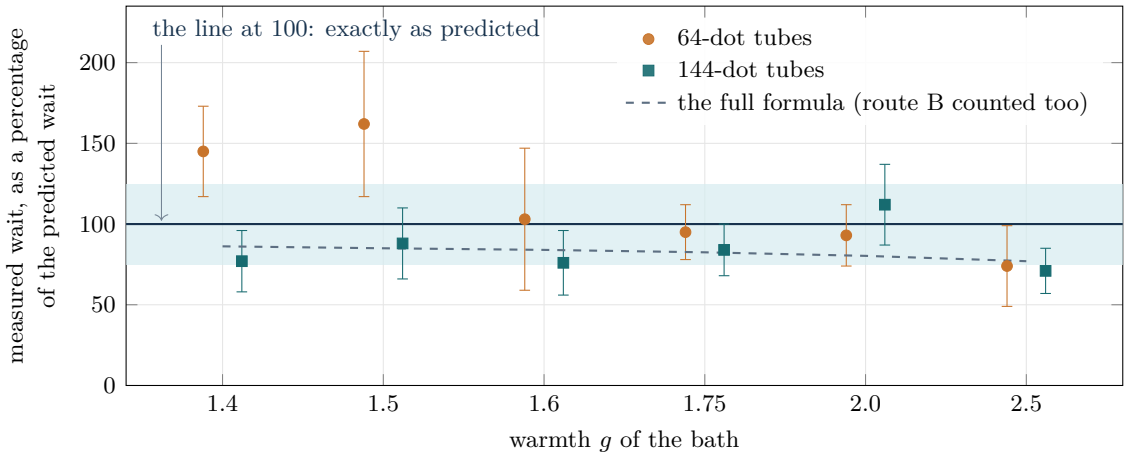


Figure 16. Twelve conditions against the count. MEASURED, EXPLORATORY Each mark is the average wait of sixteen tubes, shown as a percentage of the wait predicted from route A alone, so 100 means exactly as predicted. The bar through a mark is its uncertainty. The shaded band, 25 percent either way, is where such an average usually lands by chance when the prediction is right. The dashed line is where the full formula, with route B counted too, puts the prediction.

MEASURED A stricter, pre-registered test then counted every single accepted swap in 240 further runs across six settings of λ and size, and compared the first exits with the full formula, both routes. All 240 first exits were observed, and their mean arrived when the formula said it would in every cell, within the scatter expected from that many runs.

What kind of waiting

If the chance of escape is the same on every attempt, the waits are **memoryless**. It is like rolling a die until a six comes up: twenty rolls without one do not make a six any likelier on the next roll, and a tube that has waited a long time is no nearer to leaving. One sign of a memoryless wait is that the waits spread out by about as much as their average. MEASURED That check was pre-registered, and it passed at every size in the main run. MEASURED, EXPLORATORY Checked afterwards, the whole pattern of first exits fits that picture too.

MEASURED One more thing the move-by-move count revealed: the first swap out does not always lead anywhere. Think of that first swap as a door opening. About a third of the time the door closes again, and the tube goes back to being a perfect tube. Only 56 to 68 percent of first exits went on to a real change, with no trend in λ. The formula predicts when the door opens, not the departure.

A caution about the clock

OUR READING, UNVERIFIED The clock here counts attempted swaps, not seconds. One sweep offers any one neighborhood a way out about $1/N$ as often as a clock that gave every neighborhood its own fixed rate would. On that fairer clock the wait would fall as $1/N$, and a longer tube would go sooner. The toy has no time in it; which clock is "right" is not a question it can answer, and a later paper adopts the fairer one.

Why we went here. If the measured wait had disagreed with the counted lottery, something larger than a single swap would be holding the tube shut, and 12 would have been the wrong number. It agrees. The rim is the single swap, and the lottery is the whole explanation of the first exit.

The point to keep: Count the exits a sweep offers, multiply by the chance each is taken, and you predict the mean wait with nothing fitted. The tube keeps that appointment over a ninety-fold range of waits, and its waiting is memoryless, like rolling for a six. But about a third of the time the door closes again: an opened door is not a departure.

6 THE CHANGE

What takes shape on the other side

The experiment: the arrangement • the cost • the wait • **the change** • the energy • what remains • other settings

Chapter 5 was about the start: the first swap out, and whether it holds. This chapter is about what comes next: how the change spreads through the tube, and what the tube turns into. It spreads fast.

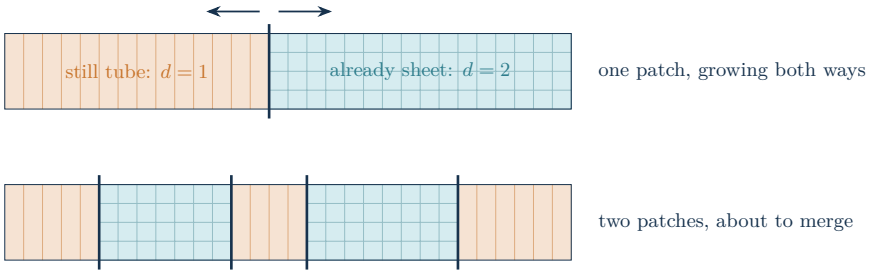
The zipper picture, and an instrument to test it

The natural picture is a zipper. One spot of the tube opens, and the opening runs along the tube the way a zipper runs along a jacket. The slider, the tab you pull, is the boundary: behind it the tube has already opened into flat sheet, and ahead of it the tube is still closed. Physicists call a moving boundary like this a **front**.

EXACT The instrument is the direction count d of Chapter 3. Dots still in the tube read $d = 1$; dots on a flat sheet read $d = 2$; a disordered dot reads 0 or 3 and up. The program marks every dot and keeps

a progress meter that runs from 0 (all tube) to 1 (all flat). It takes a snapshot of the whole network at three stages fixed in advance, so that nobody could pick a moment because its picture looks persuasive: when the meter first reads a quarter, a half and three-quarters. The predictions were scored at the half mark, where anything in between tube and sheet would have the most room to show.

MEASURED At $\lambda = 1.25$ and $g = 1.5$, thirty tubes at each of four sizes, run twice with fresh random numbers, and three predictions written down first: a memoryless wait; at least 80 percent of dots in one of the two signatures at the half mark; and at least 70 percent of the flat-like dots in one connected region, on average. All three held at every size in both runs. At the half mark 98.8 to 99.8 percent of dots read as tube or sheet. The other two snapshots, which were not scored, say the same: 98.8 to 99.7 percent at a quarter, and 98.5 to 99.5 percent at three-quarters. Something organized was happening, early, midway and late, with almost nothing in between.



The tube is drawn unrolled: four dots tall, L dots long, its ends joined. After the patches merge, the two histories leave the same final picture.

Figure 17. What "one zipper" can and cannot mean. SKETCH, NOT DATA Both histories end with one large connected sheet-like region, and a snapshot cannot always tell them apart. The tube is a loop, so even a single open patch has two sliders, one at each end, moving apart.

One zipper, or several?

A jacket found open along one long stretch could have been unzipped from one spot, or from two spots whose openings ran into each other before anyone looked. So one large open region is weaker evidence than one zipper. MEASURED The averages passed the 70-percent rule at every size, but single runs did not always. Of the thirty runs at 64, 96, 144 and 192 dots, 0, 2, 4 and 9 fell below it in the first set, and 0, 1, 3 and 5 in the second. A separate pre-registered test of a simple one-front law came out NOT ESTABLISHED: 21 of 59 tubes that started converted in several patches, mostly at the warmer bath. The honest description is that one connected flat-like region usually dominates by the half mark, and that the change often starts in more than one place.

A count is not a shape

EXACT A single swap from a perfect tube can produce dots that read $d = 2$ while their links carry one, two, two and three squares, which no dot on a flat sheet does. Two buildings can have the same number of rooms and different floor plans. So the paper calls $d = 1$ and $d = 2$ *curled-like* and *flat-like* signatures, and when it says the curled direction opened, it means only that the four-step curl is gone. Whether what is left is wide, in the sense of Chapter 3, the count cannot say. The saved wiring can.

What the tube turns into

MEASURED, EXPLORATORY The 120 final networks saved at $\lambda = 1.05$ were rebuilt, link by link, as grids

with their edges joined. All 120 are tori: one connected surface, with no twist in it. And all 120 are narrow. The shortest way around is six steps at every size tested, where the tube's was four. A second way around, independent of the first, is 16, 24, 36 and 48 steps long at 64, 96, 144 and 192 dots, as long as the tube was.

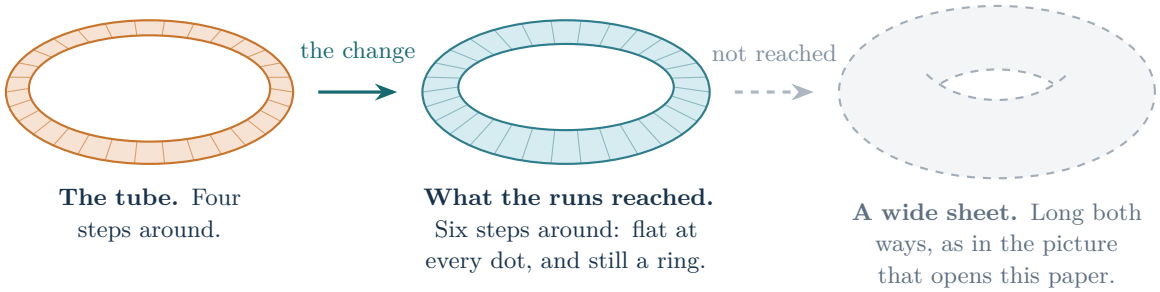


Figure 18. What the tube turns into. MEASURED, EXPLORATORY The tight curl goes and the energy comes out, but the ring only gets a little fatter: six steps around where it was four. SKETCH, NOT DATA Widths are drawn to suggest, not to scale; grey is what these runs did not reach.

How does a ring four around become a ring six around? Its seam is re-glued two places further along (Figure 19). That removes the four-step loop and leaves a six-step one. EXACT Shift the seam by two more places and the six-step loop goes as well, leaving eight; each further shift adds two. If the re-gluing kept happening, the ring would fatten into a doughnut. But the toy's price list charges for loops of four and for nothing longer. Only the first shift releases energy. The later ones are neither rewarded nor charged, so nothing pushes them along.

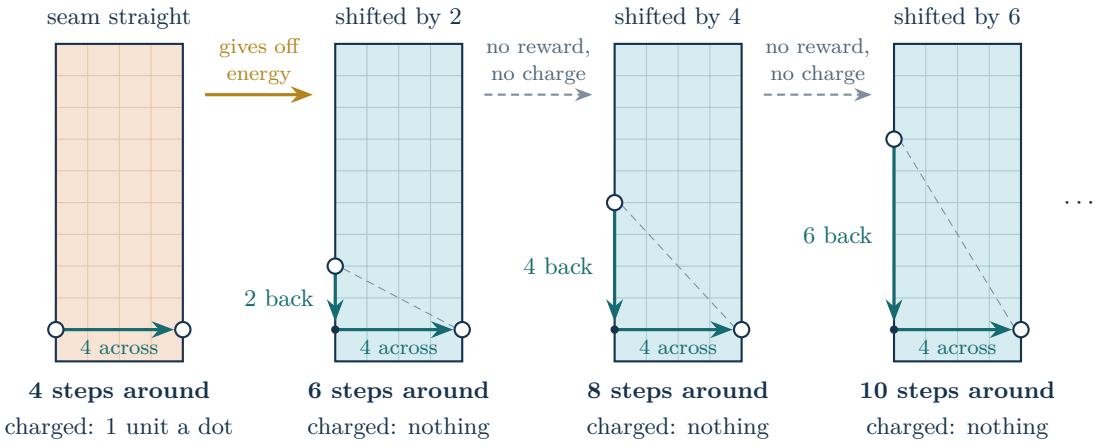


Figure 19. One shift pays; the rest are free. EXACT Each strip is a piece of the unrolled tube, four dots across. The two rings in a strip are one dot: the seam joins the end of a four-step row to it. With the seam joined straight, the row closes on itself in four steps. Each shift of the seam by two places sends the row's end two dots further along, so the shortest way around grows by two: six, eight, ten. Only loops of four are charged (one unit a dot at the setting most runs use), so the first shift gives off energy, and the later ones neither cost nor repay anything. The widths were checked up to twelve around, on tubes of 96 dots or more.

Does being flat at every dot at least make a wide shape likely? EXACT Counted as ways of wrapping the grid, yes: of the wrappings of 192 dots that are flat at every dot, about one in twelve is six around, and the typical one is twelve around. MEASURED, EXPLORATORY But the runs do not choose among them

at random. Every move out of such an arrangement costs energy, at least 20 units from a ring six around and 32 from a wider one at the setting most runs use. So in a cool bath the network stays at the first flat arrangement it reaches, and from the tube that is the nearest one: six around, where an inhabitant running the count of Chapter 3 would find the way around at three steps.

MEASURED, EXPLORATORY The newer map of Chapter 9 saved many more end states: 3,521 at zero energy, from runs that passed the three-quarter mark. Of those, 3,446 are narrow tori six around, 73 came apart into separate surfaces, and two are Klein bottles, closed surfaces with a twist in them. So "zero energy," "flat" and "one torus" are three different statements.

The energy comes out exactly

MEASURED When a tube finishes at zero energy, the energy given off is exactly one unit per dot at $\lambda = 1.25$: the arithmetic of Chapter 3, now measured. In 24 to 28 of 30 runs per size the release was within one percent of that. The rest settled on sheets with small defects, 8 to 45 units' worth read from their wiring, and held back that much. The ideal gap is exact; how much of it comes out depends on where the network stops.

Why we went here. A snapshot during the change tests whether the two signatures sit side by side and whether one connected patch dominates. Those are useful, limited questions. They do not by themselves establish a single advancing front, a wide space or a thermodynamic first-order transition.

The point to keep: A quarter, half and three-quarters of the way through, about 99 dots in 100 read as tube or sheet, and one connected sheet-like region usually dominates. That is sharp in the sense written down beforehand, and it is not proof of a single zipper. What the tube turns into is a ring six around where it was four: flat at every dot, and still narrow.

7 CLOSE THE LID

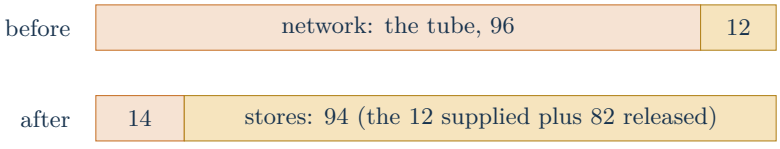
Where does the energy go when it has nowhere to go?

The experiment: the arrangement • the cost • the wait • the change • **the energy** • what remains • other settings

Apart from the budget test of Chapter 4, every run so far sat in the bath, and the bath carried the released energy away as fast as it appeared. If a change like this made all of space, there was no outside for its energy to go to. So seal the system again, as in Chapter 4, and watch what the burp does to its own surroundings.

EXACT The sealed system of Chapter 4 had one store. Here there are C stores beside the network, each holding a non-negative number of units. Every proposed swap picks one store at random: an uphill swap is refused unless that store can pay; a downhill swap deposits its release there. Nothing enters or leaves, so network plus stores is constant to the unit, and it was checked to be constant in every run.

Return to our 96-dot tube, holding 96 units. Put 12 more in a store and seal the lid. The account reads 108. Suppose the change runs and leaves one 14-unit scrap in the sheet, as Chapter 8 will show it often does. The network has gone from 96 to 14; the stores have gone from 12 to 94.



Both bars are 108 units long. Nothing is lost; it only moves between the network and the stores.

Figure 20. The ledger for the running example. EXACT The bookkeeping is exact in every sealed run; the particular ending shown here, one 14-unit scrap, is an illustration of a common outcome, not a promise about any one run.

The surroundings can spoil the ending

Here is the twist. The released energy is now in the stores, and the stores pay for uphill swaps. Energy that has nowhere to go can buy swaps that break the very squares the change has just made. How much damage depends on how many stores there are. PUBLISHED Many stores share the release thinly, like a large cold room; few stores concentrate it, like a small hot one.

MEASURED Pre-registered: tubes of 64, 96 and 192 dots, seven choices of C from a single store to two per dot, twenty runs at each setting, 420 runs. The outcomes were classified by rules fixed beforehand: sheet-like if at least 90 percent of dots read $d = 2$ with almost no melted dots; melted if 15 percent or more read as disorder; stalled if both orders sat side by side with nothing melting; in between otherwise.

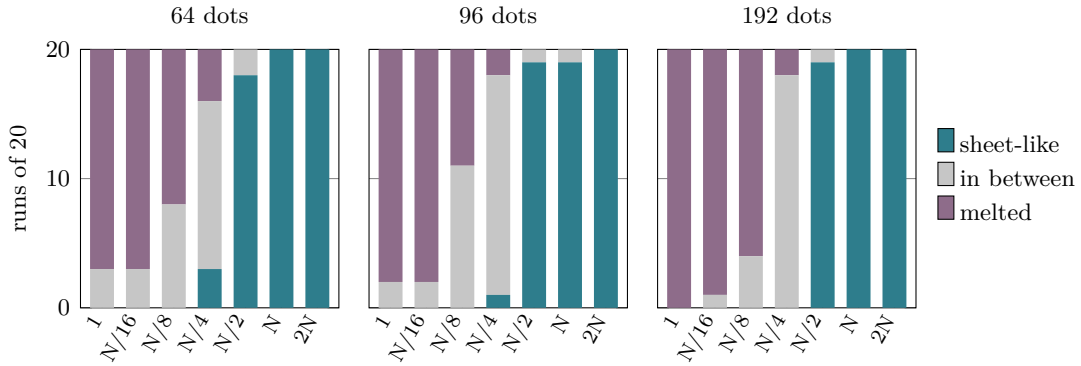


Figure 21. Bonfire or melt. MEASURED The horizontal axis is the number of stores C, as a share of the number of dots N. With few stores the release damages the new sheet (the registered label is "melted"; the next section says what that can mean); with many it leaves a clean sheet. The switch-over lies between one store per four dots and one per two at all three sizes, and no run ended stalled with both orders side by side. Totals: 179 sheet-like, 87 in between, 154 melted.

MEASURED A bonfire with a threshold, as the pre-registered verdict put it. Where the heat has room, the change feeds itself to completion; where it does not, the heat damages what the change made. The switch-over moves in step with size, as a release of one unit per dot shared among the stores predicts. OUR READING, UNVERIFIED A rough energy budget puts it near one store per three dots; the measured switch-over sits between one per four and one per two.

MEASURED No run ended stalled, with tube and sheet side by side and nothing melting, in 420 runs of 30,000 sweeps. That is an observation about these runs. It does not show that such a standoff is impossible, which would take evidence these sealed runs do not supply.

What "melted" means here

EXACT A sealed box holds only the energy it started with: at $\lambda = 1.25$, one unit per dot, plus the push. A missing square costs 16 units. So the burp's own heat can break at most about one square in sixteen, however few stores there are. MEASURED, EXPLORATORY Every one of the 154 networks classed as melted still had 94 to 100 percent of its squares. That is damage: warm fabric with flaws in it. It is not the random tangle the published model starts from, which has about an eighth as many squares as flat fabric. The burp cannot melt its own product.

Does trapped heat make the product wider?

MEASURED Pre-registered, with the author's prediction that it would: tubes of 96, 192 and 384 dots at $\lambda = 1.25$ and 1.40, sealed with two stores per dot down to one per four, then drained of their heat, cooled, and read from the wiring. It did not.

- **Plenty of room.** With one store per dot or more at $\lambda = 1.25$, every one of 144 tubes ended narrow, six around. With half a store per dot most still did.
- **Little room.** With a third or a quarter of a store per dot, or at $\lambda = 1.40$ with any amount of room, most ended as a *patchwork*: good fabric almost everywhere, six around in one stretch and eight or more in the next, with flaws frozen in where the stretches meet, and often in two or more separate pieces.
- **Either way.** None of 720 end states was one clean torus wider than six.

Use the inhabitant's count of Chapter 3: call an end state *mostly wide* if at least half its dots have 25 dots within three steps, as a dot on a wide sheet does; a dot on one of these tubes six around has 22. The share that came out mostly wide fell as the tube got longer, from one in five at 96 dots to one in sixty at 384 ($\lambda = 1.40$). For the largest size the shares were written down beforehand from the two smaller ones: they held at nine of ten settings and missed at one. OUR READING, UNVERIFIED Trapped heat is flaws in many places at once, each re-stitching its own neighborhood. A clean re-gluing needs one edge moving alone.

Why we went here. What the burp turns into, and whether the energy books balance across the change, only have meaning in a sealed system. In a bath the burp leaves and nothing can be asked of it. The sealed box is where the balance becomes a measurement.

The point to keep: Sealed, the release has to go somewhere, and where it goes decides the ending: into enough room and the sheet completes, into too little and the heat damages the new space without melting it or widening it. The books balance to the unit in every run. Nothing stalled half and half.

8 WHAT REMAINS

A scrap of the old arrangement

Almost every sheet that formed in a cold sealed box kept something: four dots of the old curled arrangement that the change swept past without opening. A remainder raises one question first: does it grow with the space? A remainder that grows in step with the space is a fixed share of everything, big enough to be some of what a universe is made of. A remainder of fixed size is a speck. The prediction, written down before the measurement, was that it grows.

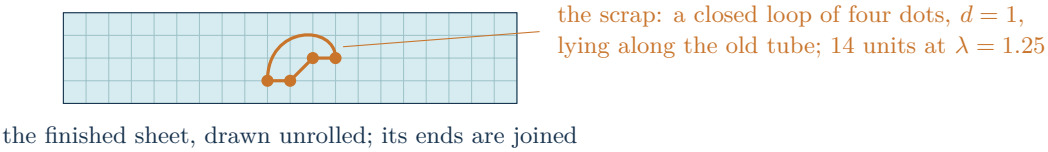


Figure 22. The usual leftover. EXACT Read from the wiring: a closed loop of four dots with one extra square and six surplus squares, costing $24\lambda - 16$ (14 at $\lambda = 1.25$), its points spanning one to four of the tube's original columns and lying along the tube. Occasionally a different, 20-unit twist appears instead.

MEASURED Eighty fresh runs in the coldest box, two stores per dot, pre-registered: tubes of 64, 96, 192 and 288 dots. The count of scraps per finished sheet was 0.90, 0.85, 1.05 and 0.95, with a spread of about 0.2 to 0.4 either way across the twenty runs at each size. A straight line drawn through them is flat, within that scatter. About one scrap per box, however long the tube.

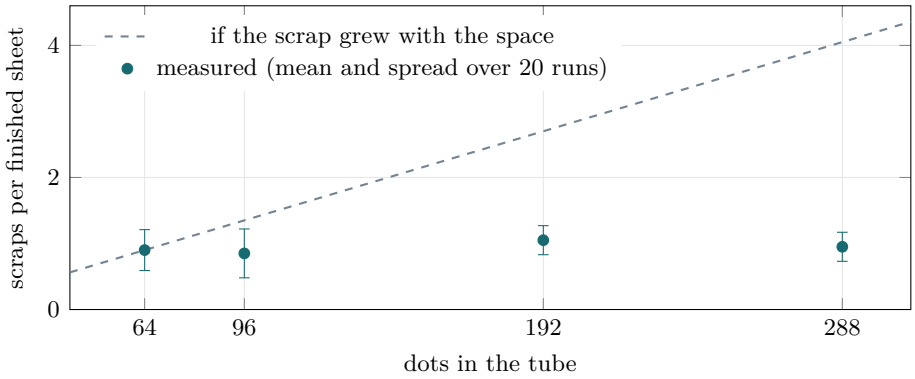


Figure 23. One scrap, however large. MEASURED The dashed line is what a leftover growing in step with the space would look like; the measured counts sit flat. The prediction that it grows, written down beforehand, failed.

Where does it sit, and why one?

It is tempting to call the scrap the place where two fronts met and left a seam. MEASURED That was pre-registered and tested: the scrap sits neither opposite the start nor at it, and a randomization check that moves it to every position along the tube finds nothing special about where it landed. We do not know why the count is about one in this protocol. The technical paper [11] says so under "not claimed."

What the scrap does afterwards

This paper stops at the count. Three questions follow from it, and they are the subject of the author's next paper: does a scrap last, or heal away, when the sheet around it is kept warm or is cooled; does a change that starts in several places at once leave more of them; and if so, is the leftover a speck or a share. The short version: how many scraps survive depends on how the change started and how fast the sheet cooled, and more than one prediction written down beforehand failed.

Why we went here. The change leaves a remainder in almost every run, at every size. The pre-registered test asked the one question that decides what kind of thing it is, and the answer, fixed size, is not the one the author predicted. It is reported as prominently as the results that went her way.

The point to keep: The change leaves about one four-dot scrap of the old arrangement per sheet, 14 units, whatever the size of the tube, sitting anywhere along it. From a single seed the leftover is a speck. What it does afterwards, and what several seeds leave, is the next paper's question.

9 THE MAP

Turning the knob

The experiment: the arrangement • the cost • the wait • the change • the energy • what remains • **other settings**

Everything so far was at one setting of the knob, $\lambda = 1.25$, chosen in advance. A single setting proves little: a result that only holds at one carefully chosen number is the kind of thing the project's rules forbid. So the whole stretch of the knob was run, from 1.05 to 1.45 in steps of 0.05, at four sizes, with the rules fixed first.

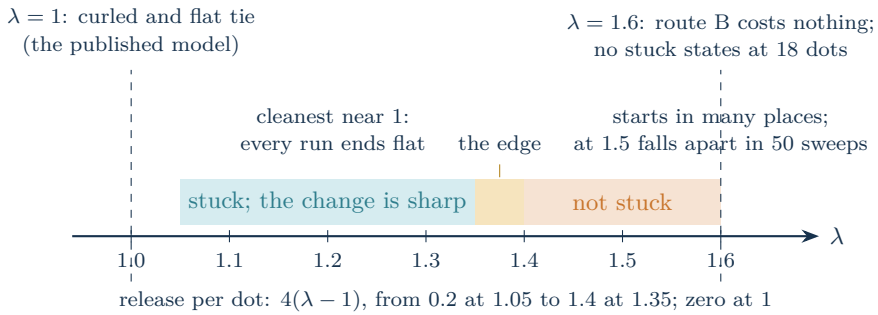


Figure 24. The stretch of the knob that was run. MEASURED Shaded, where the tube is stuck (more than half the tubes still at least three-quarters tube after 200 sweeps) and the change keeps its two-signature, dominant-region character. The edge between 1.35 and 1.40 is where the wait becomes too short to see, not where the hill disappears: route B still costs energy until 1.6. EXACT

What the map shows

MEASURED Three things.

- **Where the tube is stuck.** At every setting from 1.05 to 1.35, and at none from 1.40 up. Wherever it is stuck the change keeps the character of Chapter 6: 95 to 100 percent of dots in one signature or the other at the half mark, and one connected region holding most of the flat-like dots.

- **When it leaves.** The mean wait follows the counted lottery of Chapter 5 over a hundred-fold range of waiting times, with nothing fitted.
- **How clean the ending is.** Cleanest near the published setting, and messier as the knob turns. At 1.05 every one of 120 tubes ended as a perfect flat torus, at 1.25 about 88 percent did, at 1.35 fewer than half, and beyond the edge the change starts in several places at once in most runs and ends mostly on a sheet with defects.

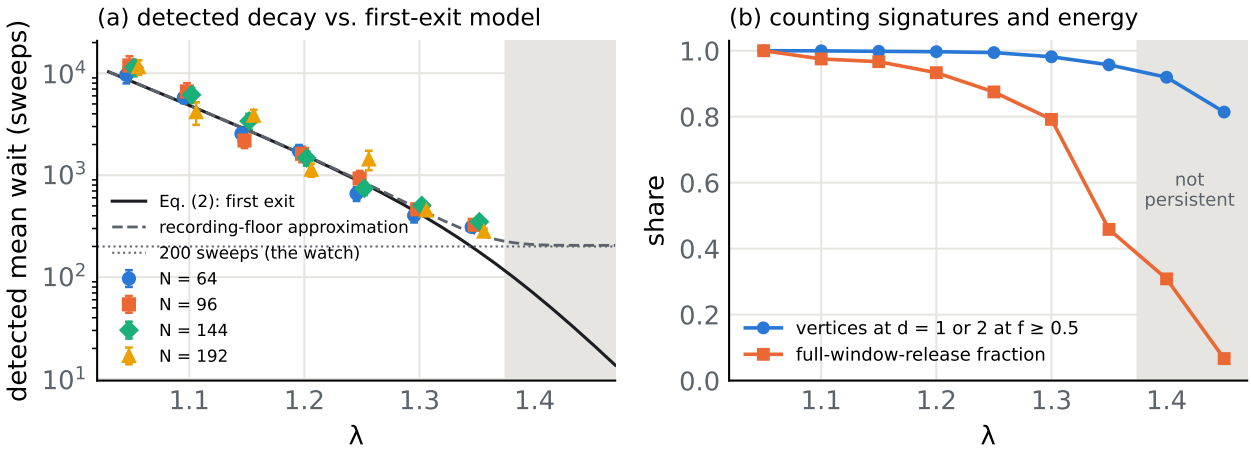


Figure 25. The map, measured. (a) Mean wait against λ at $g = 1.5$, four sizes; the solid line is the counted lottery for the first exit, and the dashed line is the same prediction for a detector that cannot see a change in its first 200 sweeps; neither is fitted. Shading marks where the tube is not stuck. (b) Pooled over sizes: the share of dots in one signature or the other at the half mark, and the share of runs that released the full amount.

OUR READING, UNVERIFIED There is a trade here. A larger λ means a larger burp per dot but a less patient X and a messier birth; a smaller λ means a cleaner change that releases less. The ideal release is $4(\lambda-1)$ per dot, reached only when the change ends at zero energy, and it vanishes at the published setting, so this family of rules has no setting with a large release and a quiet, clean X at once. PUBLISHED At the published setting itself, the published model's author has proposed that patches of the network can sit stuck in a different arrangement [6]; the questions asked here could be put to those patches too.

Does it hold up?

MEASURED The map was scored against rules fixed before it ran, and it passes: in all 24 cells from $\lambda = 1.05$ to 1.30 and 64 to 192 dots, the waits spread as a memoryless wait should, the dots sit in one signature or the other, and one connected patch dominates. A larger size, 288 dots, was predicted before it was read, and all six settings landed inside their predicted ranges. At 1.35 the connected-patch test fails at 192 and 288 dots, so the stretch where everything holds is 1.05 to 1.30.

Two things keep that pass in proportion. It took four tries: three earlier maps each failed one of their rules and stay inconclusive, for reasons that mostly traced to the measuring and not to the tube, and the technical paper [11] has that story. And the single patch weakens toward the edge: at 288 dots and $\lambda = 1.30$ the largest patch holds 75 percent of the flat-like dots on average, against a cutoff of 70, and only 65 of 120 single runs clear it.

Why we went here. The project's first success condition is no cherry-picking: a family of rules may be explored,

but only along knobs fixed in advance, and every setting run is published. The map is that rule applied, and it also locates where the stand-in for X stops being stable for now.

The point to keep: The tube is stuck, and the change keeps its character, from 1.05 to 1.35, and the map passes its registered tests from 1.05 to 1.30, including a size predicted in advance. A bigger release comes with a messier change. None of this is a test of a thermodynamic first-order transition, and the finished tori are still narrow.

10 THE EDGES

Where the simulations stop, and what lies past each edge

Every knob the toy has was turned only so far. This chapter is the honest map of the box: for each knob, what was run, why it stopped there, and what is on the other side. "Unknown" means no one has looked.

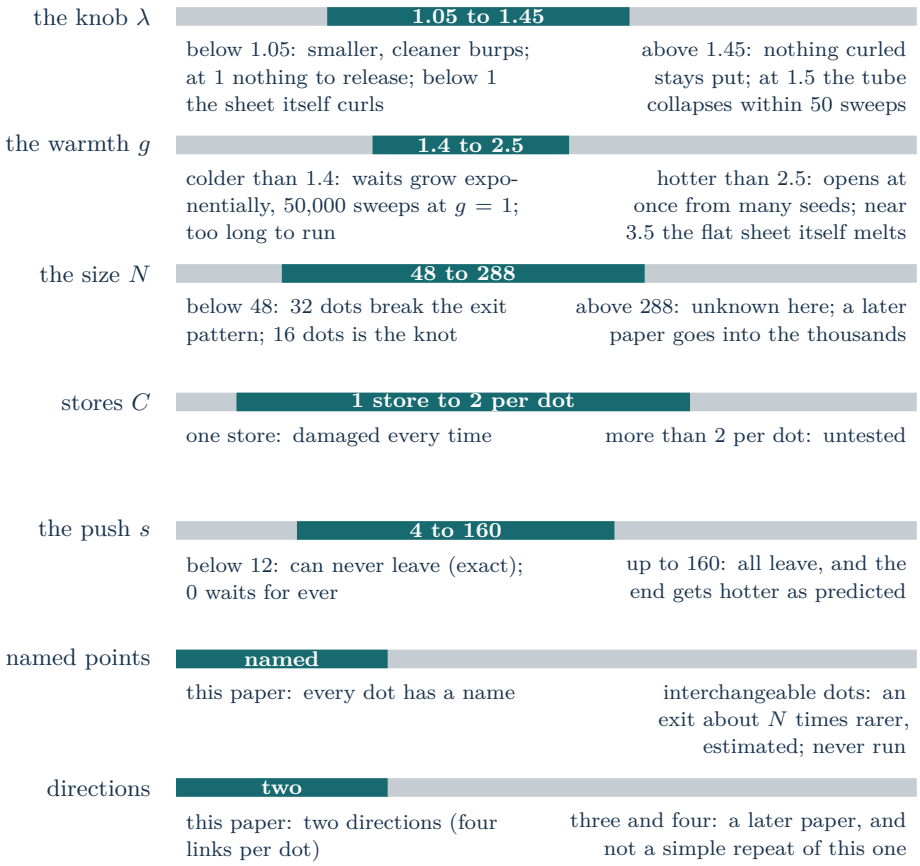


Figure 26. The box, knob by knob. Each teal bar is the stretch of a knob where tubes were actually simulated for this paper; the gray is the knob's range as far as anything is known about it; the notes say what lies past each edge.

You can turn the first five knobs yourself in the interactive companion to this figure, The Tube's Curling Ladder: its exact numbers follow you to any setting, and its measured results appear only inside the teal bars.

The knob λ: 1.05 to 1.45

Below 1.05 the burp gets smaller and cleaner, not longer-awaited. EXACT The wait levels off as the knob approaches 1 (the counted lottery gives about 8,300 sweeps at 1.05 and about 14,000 at 1.001), while the release shrinks to nothing at exactly 1, where curled and flat tie. Below 1 the ladder flips. MEASURED, EXPLORATORY In a rehearsal the author calls the backwards reaction, a perfect flat sheet at $\lambda < 1$ sat at zero for hundreds to thousands of sweeps and then curled, one direction first, releasing energy in steps that landed on the exact rungs of the ladder. MEASURED Above the range, at 1.5, the tube falls apart within 50 sweeps in every run. EXACT At 1.6 route B costs nothing, and at 18 dots no stuck state above flat exists; by a setting of 5 to 10 the toy behaves as one with a third square simply forbidden. Nothing curled is stable for now out there.

The warmth g: 1.4 to 2.5, with the maps at 1.5

EXACT Colder, the wait grows like $e^{12/g}$: about 50,000 sweeps at $g = 1$ and nearly three million at $g = 0.75$. MEASURED, EXPLORATORY At $g = 1$, two of four tubes were still tubes after 30,000 sweeps, which is why the range stops there. Hotter, the tube opens almost at once and from many places, and near $g \approx 3.5$ the flat sheet itself melts into the random tangle; the reservoir estimate of Chapter 7 used that melting point. How it moves with the size of the sheet is taken up in a later paper, and until then every warm result carries a size caveat.

The size N: 48 to 288 dots

The upper limit was computing time, spent on repeats rather than size. EXACT The exit count was certified at 48 through 288; a control at 32 dots breaks the pattern, and 16 dots is the knot. MEASURED The front's speed fell roughly as the size to the power -0.7, so longer tubes take longer to finish. OUR READING, UNVERIFIED Whether the wait itself grows, stays flat or falls with size depends on the clock, which the toy does not fix (Chapter 5). The technical paper claims nothing beyond 288; networks into the thousands of dots are the subject of a later paper on the scrap.

The sealed box: stores and push

MEASURED With a single store every tube ends damaged; two stores per dot leave clean sheets with one scrap. Colder boxes than that were not tried. EXACT With a push below 12 nothing can leave by single swaps at the certified sizes, and a tube with nothing to spare waits for ever. MEASURED, EXPLORATORY Pushes of 12 to 160 units at 64 dots all left, and the hotter the box the more disordered the sheet it ended with, as predicted from the toy's equilibrium curve before the run: 0.97 squares per dot measured against 0.95 predicted at the smallest push, 0.81 against 0.81 at the largest. Larger pushes were not run.

Named dots, two directions, one starting point

OUR READING, UNVERIFIED Every run treats dot 17 as a different thing from dot 41. Giving each distinct wiring one weight, whatever its dots are called, would be a different way of counting, and quantum indistinguishability does not by itself require it. Energies and the certified 12-unit bound would stay the same. For the counted route-A exits at 64 and 96 dots the chance of acceptance would fall by a factor between $N/2$ and N, which does not settle the whole waiting time: other exits, and the time spent in arrangements of equal energy, matter too. This paper does not simulate it; that knob is decided on

and not yet turned.

On ideal grids, four links per dot give two directions; six and eight give three and four (Figure 10), though the number of links alone does not give an arbitrary network a dimension. Those are the subject of a later paper. The tease: a curled direction in three dimensions does not open the way the tube does, and the reason is partly exact and partly still open.

MEASURED And the starting point matters. The published route starts from a random tangle; this paper starts from an ordered tube. Both were run. From randomness at $\lambda \geq 1$: no sign of a release at these sizes. From the tube at $\lambda > 1$: a hill, a push, a burp. They are different questions, and the toy gives them different answers.

The point to keep: The box is small and its edges are known. Past most of them lies either an exact statement (nothing below 12 can leave; nothing is below flat), a later measurement with its own paper, or a plain "unknown." The one edge the toy cannot push on its own is the clock.

11 WHAT WE HAVE LEARNED, AND WHAT COMES NEXT

A mechanism we can price, and two questions it opens

Return one last time to the 96-dot tube. Here is everything this paper asked of it, with the answer and the kind of answer it is.

The question	The answer, in the toy	Kind
What is it holding?	96 units above flat: one per dot at $\lambda = 1.25$.	EXACT
What does it cost to start?	12 units, the same at every size certified, from 48 to 288 dots.	EXACT
When does it first leave?	When the counted lottery says: over a ninety-fold range of waits in a first scan, and in every cell of a stricter registered test.	MEASURED
What takes shape?	Halfway through, 99 dots in 100 belong to one arrangement or the other.	MEASURED
What does it turn into?	A ring six around where it was four: flat at every dot, and still narrow.	MEASURED, EXPLORATORY
Where does the energy go?	Finishing flat, it gives off exactly one unit per dot. Sealed, every unit is followed into the stores.	MEASURED
What remains?	About one four-dot scrap, 14 units, whatever the size of the tube.	MEASURED
Does it hold at other settings?	The fourth map passes from 1.05 to 1.30. Three earlier maps stay inconclusive.	MEASURED

The rows come from different tests and different ranges of settings; the sealed ones are at $\lambda = 1.25$ only.

What the toy shows, and where it stops

OUR READING, UNVERIFIED The toy gives the larger hypothesis a concrete first question: can a prepared curled arrangement wait, escape and release an amount of energy that can be calculated? Under the stated rules, it can. But its ending limits the comparison with the picture that opens this paper: the runs stop at the nearest flat arrangement, the narrow one, and their own trapped heat does not widen it. Two cautions go with the result. The fixed cost is the cost of opening the door, not of committing to the change. And an energy gap between two ideal arrangements is not a test of a thermodynamic first-order transition.

THE AUTHOR'S CLAIM The author's wider proposal is that an organized arrangement came before space. This paper is a controlled test of part of that proposal in a toy, not a test against observations of the universe. What it hands to the next model: the local energy can be priced exactly, the first exit can be predicted, and flatness dot by dot is not enough to make a shape wide. What did not go the way it might have: the remainder did not grow, a single front was not established, and three registered maps stayed inconclusive before a fourth passed.

What comes next

The question that began the work survives its first test. Can an organized arrangement give way sharply, release energy and leave a remainder? In this small laboratory it can, and the first step can be priced to the unit. Three things were left standing at the edge of what this paper measured, and each has work of its own coming, with its predictions written down first.

The scrap. A scrap is the only thing in the new space that still carries the old arrangement, so the next paper, *What the burp leaves behind*, asks what it takes to keep one. MEASURED One result: the scrap counted here is not permanent. Cool the new sheet slowly enough and it heals away, while another kind of leftover survives every cooling we could run.

Three directions. The tube has one long direction, and space has three, so the paper after that, *How curled directions open*, runs the same rules with six links per dot. MEASURED One result: escape from the four-link tube does not settle the question for more directions. Later runs find a separate barrier for each curled direction. Some end in wider shapes, but no fixed budget has been shown to open every direction.

A rule that pays for width. The toy's rule charges only for loops of four, which is why the first re-gluing is paid for and the next is not. So the author asked for the same rule one loop-length up: a small price on loops of six, which a ring six around has many of and a wide sheet has none. MEASURED What happened, at 96 and 192 dots, with a larger size still running. With no such price the tubes ended six around, as in this paper. With one, they did not stop there: a ring six around gave way without any push, a curled tube went straight past six around, and at three prices of the four tried most runs ended mostly wide. At the gentlest price most tubes waited out the run or ended six around. The wide result is not a clean sheet. Flaws are frozen in all through it, so that nine dots in ten have one within three steps. The price is our addition and not part of the published model.

OUR READING, UNVERIFIED Two things follow for the papers above. For the scrap: once the tube widens, leftovers stop being a rare remainder and become the main obstacle, so that paper has to ask how many are made and whether slow cooling removes them. For three directions: a direction does not become long just because its tightest loop is gone. Something has to pay for each further step, and those runs

should measure how far around an opened direction really is.

Twelve units start an escape. What opens a space?

Continuing into the technical paper

This edition follows the technical paper [11], which holds the equations, tables, uncertainty calculations and full references. Every number above is taken from it or from the project's record, and the kinds-of-statement tags follow the project's own rule that no statement is promoted from one kind to another without a dated note saying why. Its sections run in the same order as the chapters here: the toy and its exact results (Chapters 2 and 3), the cost of leaving (4), the first exit (5), the conversion (6), the sealed system (7 and 8) and the dependence on λ (9).

A few translations for the next step

In the paper	What to keep in mind
"The model"	In the technical paper, our toy: the published model with the knob λ. Only $\lambda = 1$ is the published model, combinatorial quantum gravity.
N	Number of dots. The running example uses $N = 96$.
Torus	Any grid that wraps both ways: the surface of a doughnut. The flat torus is the sheet (a fat doughnut), the curled torus is the tube (a thin ring), and the smallest one, curled both ways, is the knot.
Flat, wide	Flat is about each dot's surroundings: every link on exactly two squares, at zero energy. Wide is about the whole shape: many steps around both ways. A torus six around is flat and narrow.
$4 \times L$	Four steps around the short wrap and L around the other, L even and above 4. 4×4 is the knot.
λ **and** g	Lambda sets the price of a surplus square and so the whole ladder; g is the bath's warmth, which sets how often an uphill swap is accepted.
$H = 16(N - S) + 4\lambda X$	The price list: S squares, X surplus squares. Zero on the flat sheet; $4(\lambda - 1)$ per dot on the tube.
$\Delta E^{\ddagger} = 32 - 16\lambda$	The rim: the cost of route A, 12 at $\lambda = 1.25$. Route B costs $64 - 40\lambda$.
d	The square-count signature: pairs of links at a dot that close no square. Sheet 2, tube 1, knot 0. Not a certified geometry.
f	The progress meter, 0 for the tube and 1 for the sheet. Disorder raises it too.
Arrhenius law	Uphill events becoming much rarer as the bath cools, as $e^{-\text{cost}/g}$. Here its two ingredients are counted, not fitted.
Memoryless	The chance of leaving is the same on every attempt, like rolling a die until a six comes up; a long wait so far makes leaving no nearer.
Metastable	Stable for now: escaping the tube's neutral family costs energy. The map's edge near 1.37 is where the wait becomes too short to see, not where that stops being true.
Microcanonical	Sealed: total energy fixed. Inferring temperature from the stores also requires a checked equilibrium distribution.
Pre-registered	The prediction and the rules of the test were committed before the run. Exploratory runs say so.
95% interval	A range from a statistical procedure that, under its assumptions, covers the truth 95 times in 100.

Where the evidence is recorded

The technical manuscript and its methods supplement are in the recorded repository version, and the supplement maps every claim to code, configurations and result files. Experiment labels in that record: T7 and T8 (the decay and the map), T9 (the sealed reservoir), T10 and T11 (the scrap and its position),

T22 (exits counted move by move), T38 (the long-wait test). The questions teased above, what the scrap does afterwards, larger networks and more directions, have their own labels in the record and their own papers. The push threshold covers 48 to 192 dots and the exit enumeration 48 to 288; neither extends to other sizes.

Disclosure

The author used Claude (Anthropic) and ChatGPT/Codex (OpenAI) to assist with code, analysis and writing, and is responsible for the result. This edition was drafted with Claude from the technical manuscript and the project's record. Its analogies and pictures introduce no new experiments or evidence. No physicist has reviewed any of it.

Sources

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[4] A. Gorsky and O. Valba, *Interacting thermofield doubles and critical behavior in random regular graphs*, Phys. Rev. D **103**, 106013 (2021), arXiv:2101.04072.

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[11] E. Smith, *The curled torus burps: activated escape and conversion in a graph model of emergent geometry*, technical manuscript, 11 October 2026, in the project repository.

[12] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[13] The written search behind this sentence, 10 October 2026: where we looked, with what terms, and what came closest. File `docs/reading/notes/2026-10-10_prior_work_lambda_above_one.md` in the project repository.

[14] C. Kelly and C. A. Trugenberger, *Combinatorial quantum gravity: emergence of geometric space from random graphs*, arXiv:1811.12905 (2018).

[15] Planck Collaboration, *Planck 2015 results. XVIII. Background geometry and topology of the Universe*, Astron. Astrophys. **594**, A18 (2016), arXiv:1502.01593. Described here from its abstract; we have not read it in full.

Sources 1 to 10 carry the same numbers as in the technical paper [11]; 11 to 15 are this edition's own.